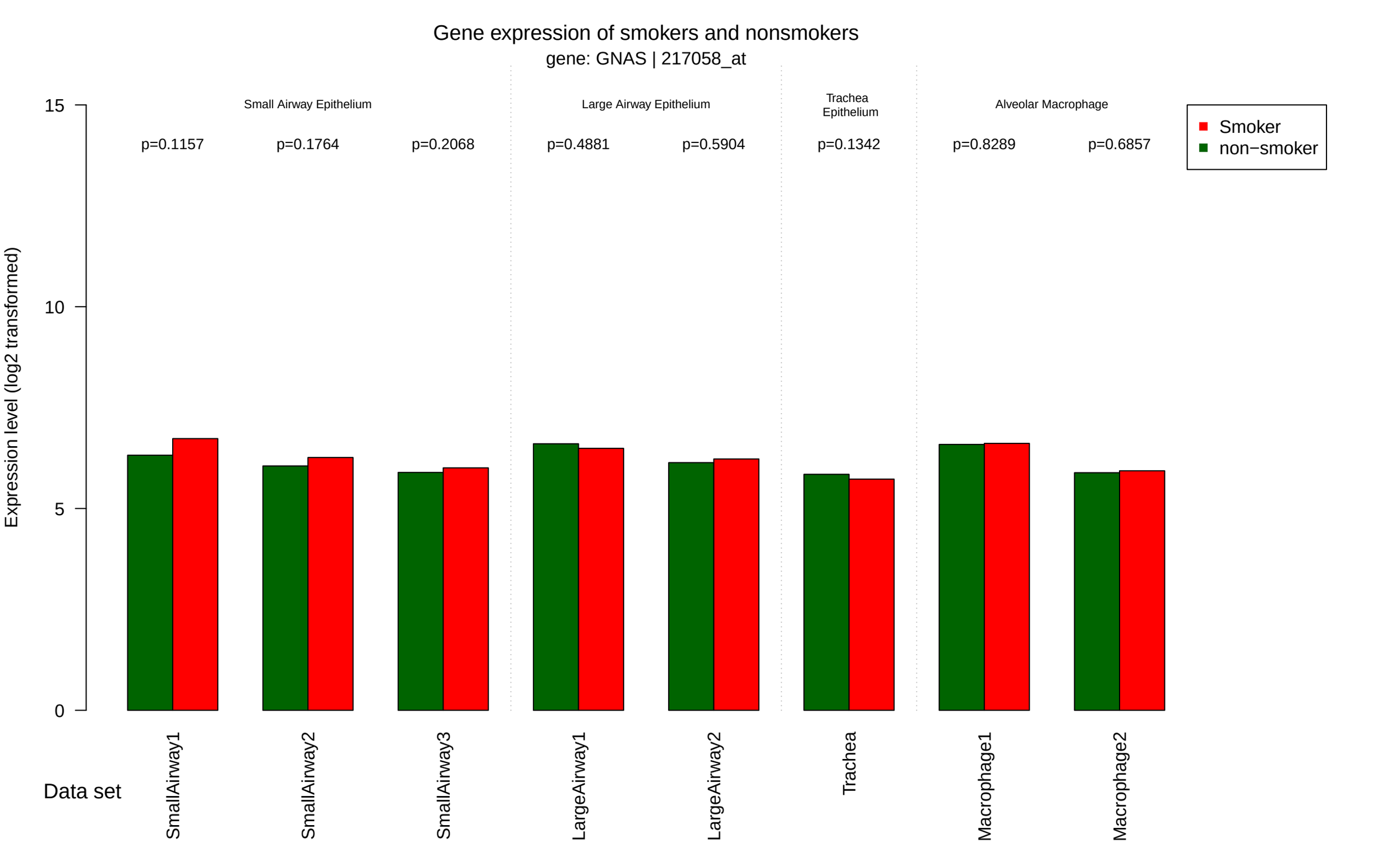
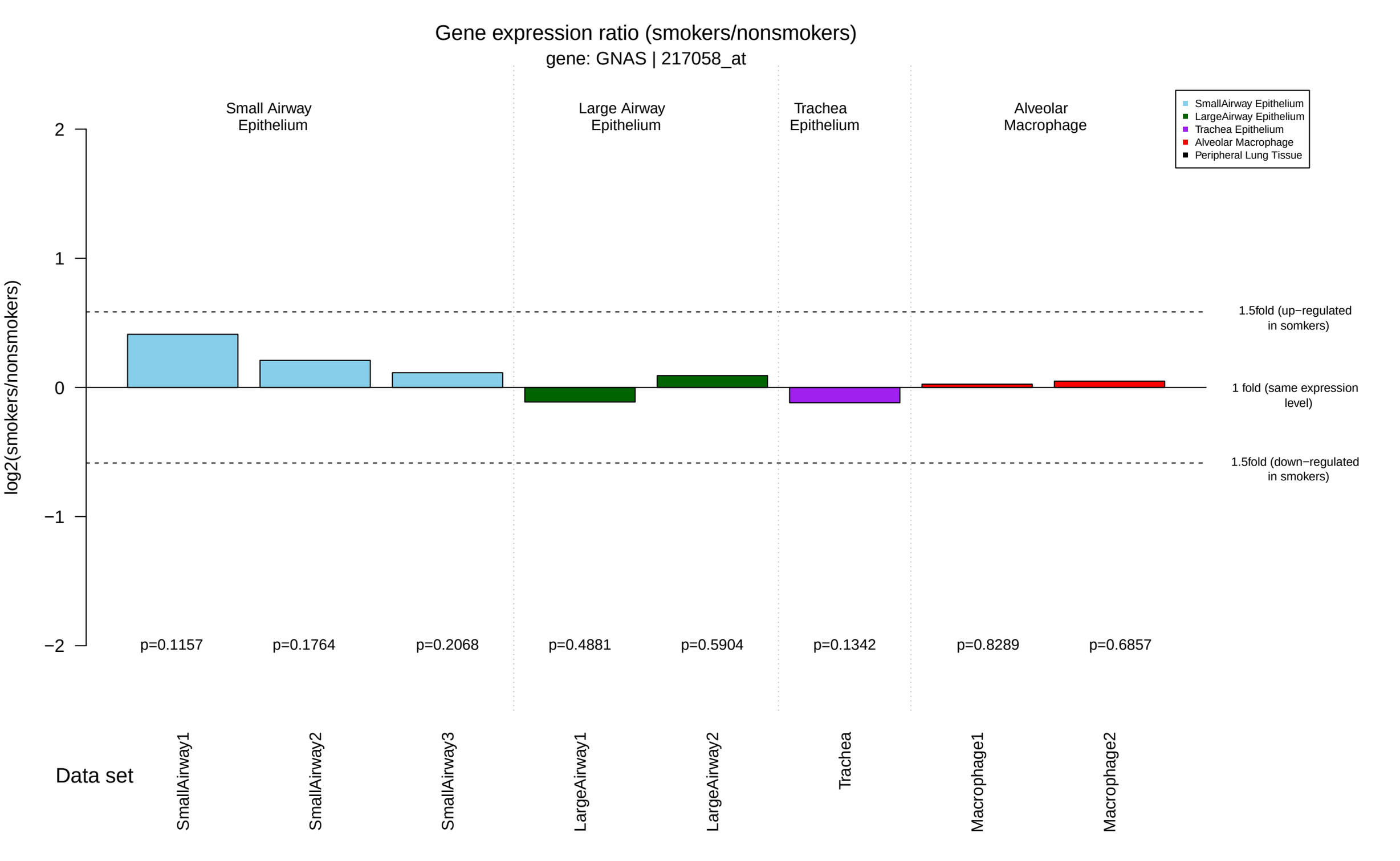
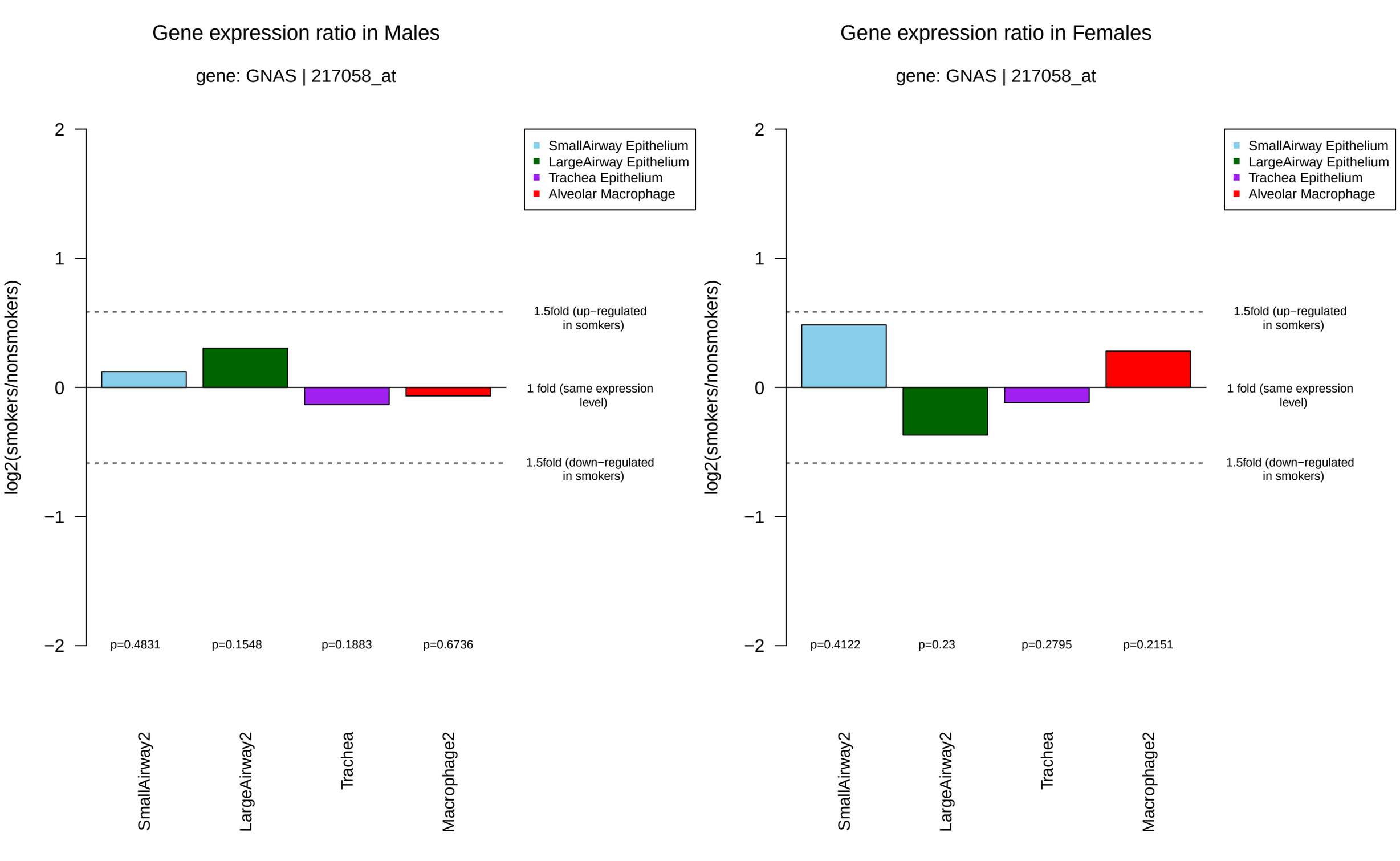
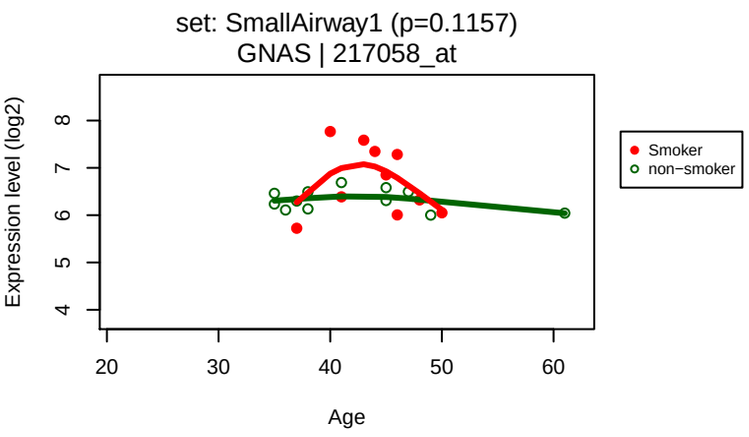
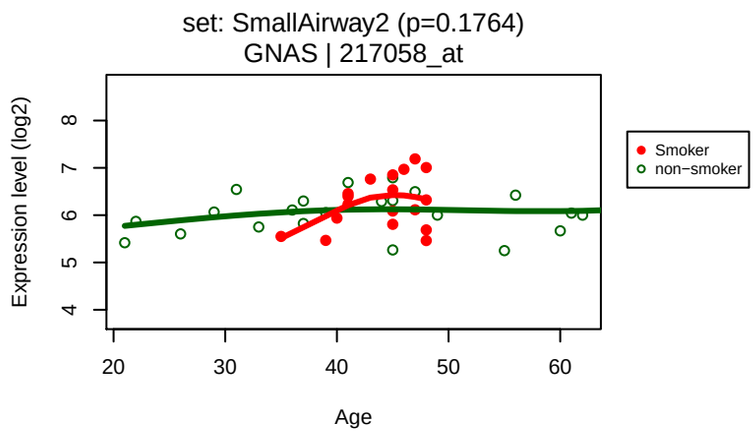
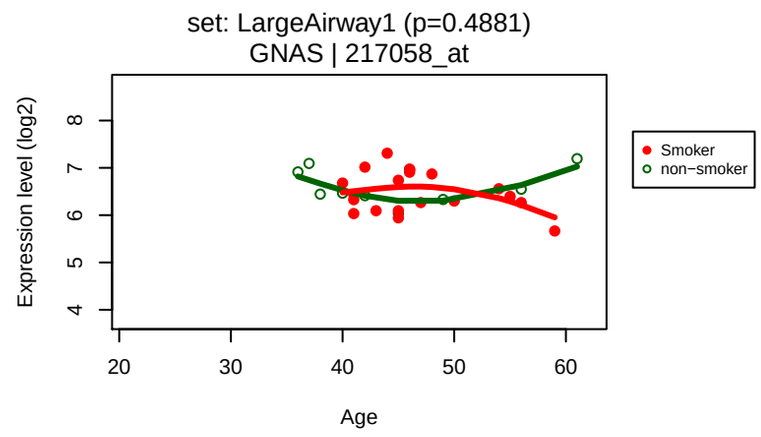

Gene expression of smokers and nonsmokers
gene: GNAS | 217058_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1157
p=0.1764
p=0.2068
p=0.4881
p=0.5904
p=0.1342
p=0.8289
p=0.6857
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

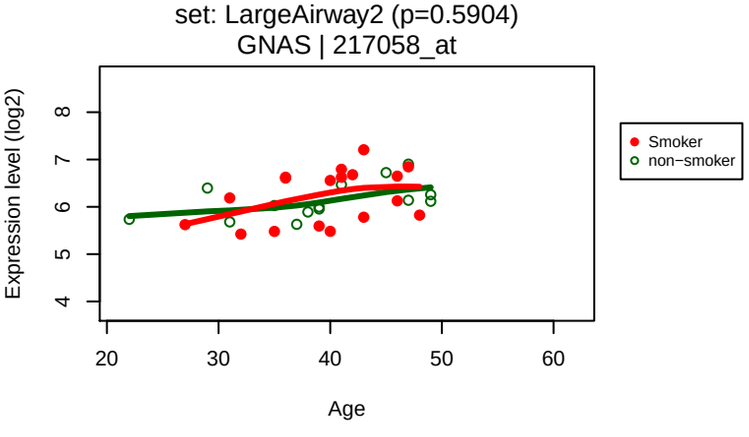

Gene expression ratio (smokers/nonsmokers)
gene: GNAS | 217058_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1157
p=0.1764
p=0.2068
p=0.4881
p=0.5904
p=0.1342
p=0.8289
p=0.6857
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

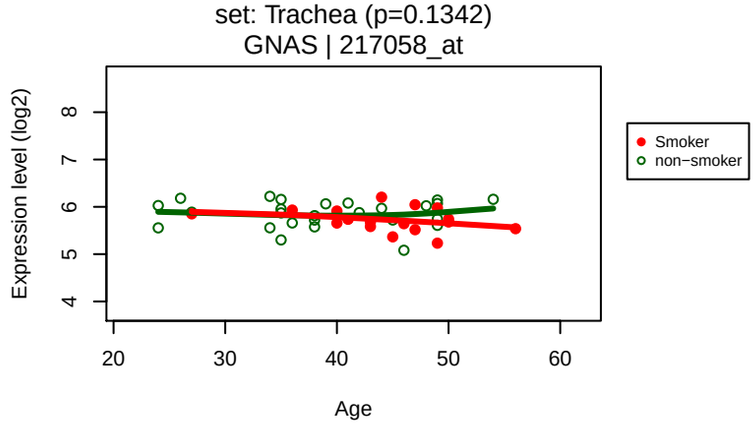

Gene expression ratio in Males
gene: GNAS | 217058_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4831
p=0.1548
p=0.1883
p=0.6736
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAS | 217058_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4122
p=0.23
p=0.2795
p=0.2151
SmallAirway2
LargeAirway2
Trachea
Macrophage2

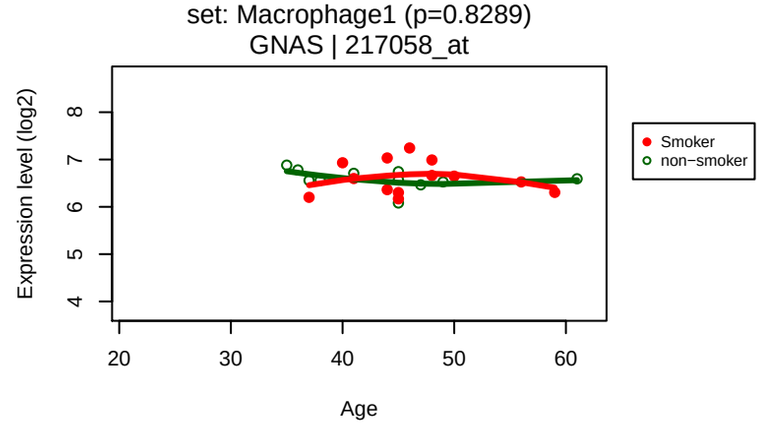

set: SmallAirway1 (p=0.1157)
GNAS | 217058_at
Expression level (log2)
Age
Smoker
non−smoker

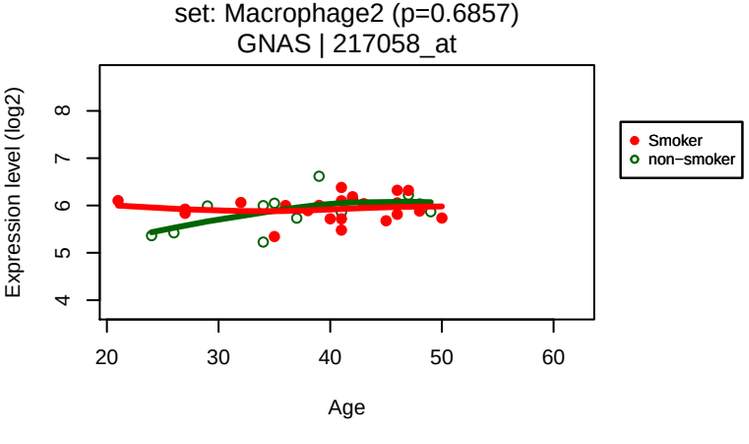

set: SmallAirway2 (p=0.1764)
GNAS | 217058_at
Expression level (log2)
Age
Smoker
non−smoker

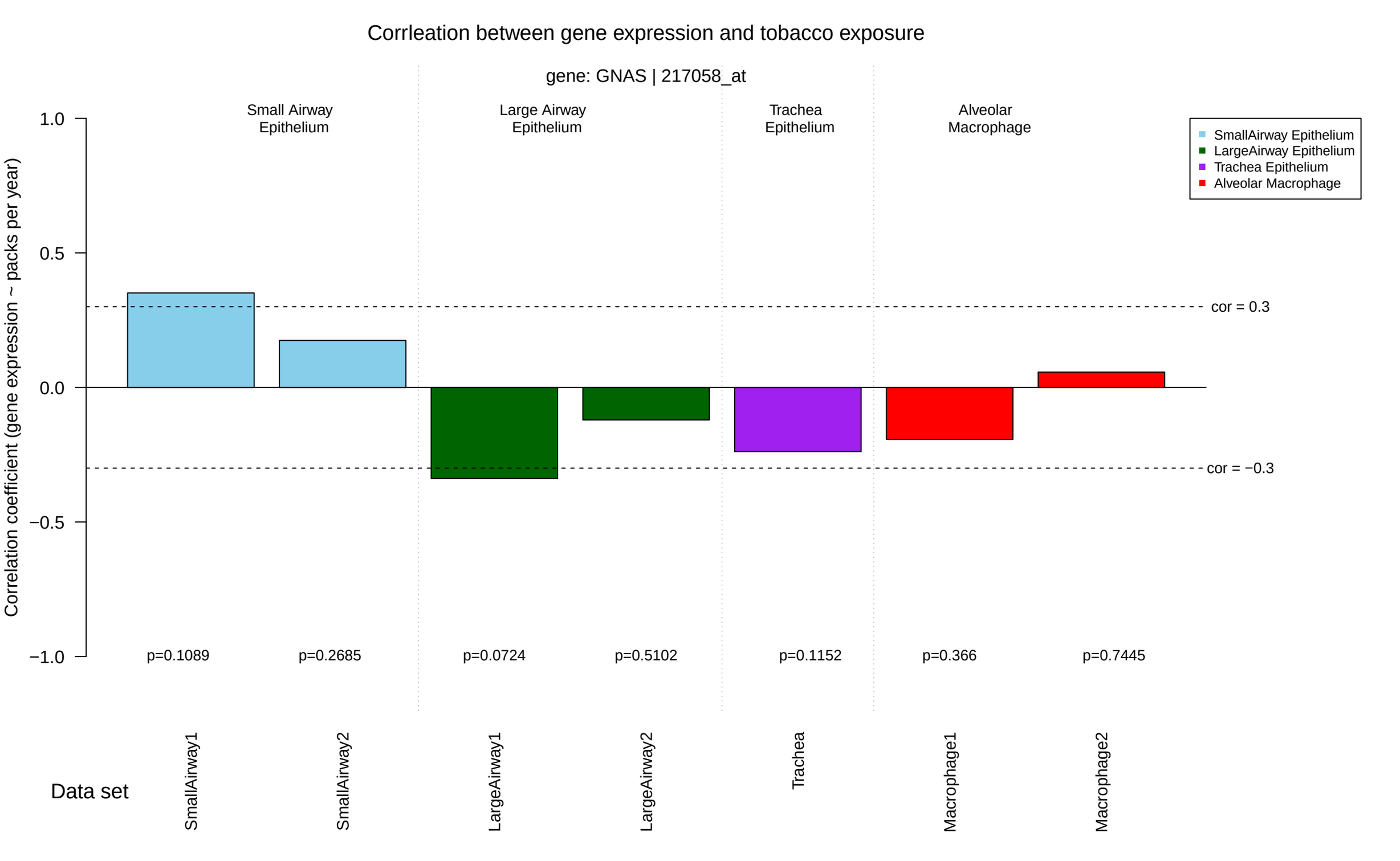

set: LargeAirway1 (p=0.4881)
GNAS | 217058_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5904)
GNAS | 217058_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1342)
GNAS | 217058_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8289)
GNAS | 217058_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6857)
GNAS | 217058_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAS | 217058_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1089
p=0.2685
p=0.0724
p=0.5102
p=0.1152
p=0.366
p=0.7445
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2